

REACHING COMMON KNOWLEDGE—A BRIEF NOTE ON A THEOREM BY GEANAKOPLOS AND POLEMARCHAKIS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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The purpose of this note is to present an illustrative example on the theorem from John Geanakoplos and Herakles Polemarchakis that shows that repeatedly communicating posteriors leads to agents to converge to a common posterior, which then becomes common knowledge. This notes completes my discussion on beliefs during Fall 2025.

1 The theorem

Theorem (Geanakoplos & Polemarchakis, 1982). Let $(\Omega, \mathcal{F}, \mu)$ be a probability space. Fix $A \in \mathcal{F}$. Let X_1 and X_2 be two random variables on this space. Let $Q_1^0 = \mu(A | X_1)$ and $Q_2^0 = \mu(A | X_2)$. For $t = 1, 2, \dots$, let

$$Q_1^t = \mu(A | X_1, Q_2^0, Q_2^1, \dots, Q_2^{t-1}), \quad Q_2^t = \mu(A | X_2, Q_1^0, Q_1^1, \dots, Q_1^{t-1}).$$

Then $\lim_t Q_1^t$ and $\lim Q_2^t$ (almost-surely) exist and are equal.

1.1 Discussion of the Assumptions

I highlight some of the assumptions underlying the theorem:

1. We work in a setting with a *common prior* μ over the probability space $(\Omega, \mathcal{F}, \mu)$. This ensures that differences in beliefs among agents arise only from differences in their private information, not from differences in their initial probability assessments.
2. The random variables X_1 and X_2 play the role of signal functions in the belief-space framework. Each agent observes the realization of their signal, namely $X_1(\omega) = x_1$ for player 1 and $X_2(\omega) = x_2$ for player 2.*
3. The initial posteriors, $Q_1^0 = \mu(A | X_1)$ and $Q_2^0 = \mu(A | X_2)$, are themselves random variables. Their realizations, q_1^0 and q_2^0 , represent each agent's first posterior estimate upon seeing only their private signal.
4. The sequences $\{Q_1^t\}_{t=0}^\infty$ and $\{Q_2^t\}_{t=0}^\infty$ represent the updated posteriors after iterative rounds of communication. At each step, each player conditions on their own signal together with the sequence of communicated posteriors of the other agent. The realizations q_1^t and q_2^t are the specific posterior estimates reached in state ω after t rounds.
5. The theorem establishes that the limits $\lim_{t \rightarrow \infty} Q_1^t$ and $\lim_{t \rightarrow \infty} Q_2^t$ exist and coincide almost surely.† In other words, through infinite communication of posteriors, agents converge to a *common posterior*, which then becomes *common knowledge*.

*Formally, these random variables induce *self-evident algebras*: the σ -algebras generated by the inverse images of measurable sets under X_i . These algebras encode the partitions of the state space corresponding to what each player can distinguish. I keep this remark as a footnote, since it requires more advanced measure-theoretic machinery.

†“Almost surely” means that the set of states $\omega \in \Omega$ where convergence fails has probability zero under μ .

6. In the special case where the state space Ω is finite, the process does not require infinitely many iterations: the common posterior is attained after finitely many rounds of communication.

The interpretation of the theorem is that each posterior encodes partial information about the other agent's private signal. When an agent communicates their posterior, they are indirectly revealing something about what they observed. With repeated exchanges, all this information is gradually revealed: after finitely many rounds in the finite case, or in the limit in the infinite case, nothing remains hidden. At that point, both agents' beliefs coincide, and this agreement itself becomes common knowledge.

2 Finite Case

In the finite case, we can describe a process of repeated information transmission that makes clear how convergence arises. At the start, each agent publicly announces the conditional probability they assign to the event A given their information. Once these announcements are made, each agent can eliminate some states of the world as impossible, namely those in which the other player would have assigned a different conditional probability than the one just announced. With this refinement of the state space, each agent updates their own conditional probability of A and publicly announces the revised value. After hearing the new announcements, the players again eliminate states in which those announcements would not have been made, update their conditional probabilities, and announce them once more. This process can be repeated as many times as necessary. Because the state space is finite, the refinement stops after finitely many rounds: the players converge to a common conditional probability of A , and this value becomes common knowledge.

2.1 Example

Consider the following Belief Space:

- $N = 1, 2$.
- $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$.
- $T_1 = \{a, b\}, T_2 = \{a', b'\}, \tau_1(\omega_1) = \tau_2(\omega_2) = a, \tau_1(\omega_3) = \tau_1(\omega_4) = b, \tau_2(\omega_1) = \tau_2(\omega_2) = \tau_2(\omega_3) = a', \tau_2(\omega_4) = b'$.
- The prior is given by $\mu(\omega_1) = \mu(\omega_4) = 1/3, \mu(\omega_2) = \mu(\omega_3) = 1/6$.

Firstly, note that the types and signaling functions generate the following information partitions:

$$\mathcal{P}_1 = \{\{\omega_1, \omega_2\}, \{\omega_3, \omega_4\}\}, \quad \mathcal{P}_2 = \{\{\omega_1, \omega_2, \omega_3\}, \{\omega_4\}\}.$$

Now consider the event $A = \{\omega_2, \omega_3\}$. As mentioned in the previous section, the signaling functions τ_1 and τ_2 play the role of the random variables X_1 and X_2 in the theorem.

I will now trace the dynamic process described above.

- Player 1 and Player 2 first compute their posteriors on A :

$$Q_1^0(\omega) = \begin{cases} \frac{1}{3} & \omega \in \{\omega_1, \omega_2\} \\ \frac{1}{3} & \omega \in \{\omega_3, \omega_4\} \end{cases}, \quad Q_2^0(\omega) = \begin{cases} \frac{1}{2} & \omega \in \{\omega_1, \omega_2, \omega_3\} \\ 0 & \omega = \omega_4 \end{cases}.$$

- Player 1 and Player 2 announce their posteriors, and they recompute them after the announcement. Notice that any announcement of Player 1 does not provide new information, since he always gives the same estimate irrespective of the state. On the other hand, the announcement of Player 2 gives new information to Player 1: if the state were ω_3 he would have heard one value, and if it were ω_4

he would have heard another. This allows Player 1 to distinguish these states and refine his partition. Thus, the new posteriors are:

$$Q_1^1(\omega) = \begin{cases} \frac{1}{3} & \omega \in \{\omega_1, \omega_2\} \\ 1 & \omega = \omega_3 \\ 0 & \omega = \omega_4 \end{cases}, \quad Q_2^1(\omega) = \begin{cases} \frac{1}{2} & \omega \in \{\omega_1, \omega_2, \omega_3\} \\ 0 & \omega = \omega_4 \end{cases}.$$

- Once again, Player 1 and Player 2 announce their posteriors, and they recompute them after the announcement. The announcement of Player 2 does not provide new information to Player 1. On the other hand, the announcement of Player 1 gives new information to Player 2: if the state were ω_2 he would have heard one value, and if it were ω_3 he would have heard another. This allows Player 2 to refine the corresponding partition element. Thus, the new posteriors are:

$$Q_1^2(\omega) = \begin{cases} \frac{1}{3} & \omega \in \{\omega_1, \omega_2\} \\ 1 & \omega = \omega_3 \\ 0 & \omega = \omega_4 \end{cases}, \quad Q_2^2(\omega) = \begin{cases} \frac{1}{3} & \omega \in \{\omega_1, \omega_2\} \\ 1 & \omega = \omega_3 \\ 0 & \omega = \omega_4 \end{cases}.$$

At this point their posteriors have become common knowledge (be sure to check this). This implies that further announcements would not reveal new information: from $t = 3$ onward, the posteriors remain the same as in step $t = 2$.

To see this more concretely, suppose that the true state of the world is ω_3 . Then the sequence of announcements and updates unfolds as follows:

- Player 1 and Player 2 compute $q_1^0 = Q_1^0(\omega_3) = 1/3$ and $q_2^0 = Q_2^0(\omega_3) = 1/2$.
- Player 1's announcement does not give new information to Player 2, but Player 2's announcement allows Player 1 to learn that the true state of the world is not ω_4 , because if it were, Player 2 would have ascribed conditional probability 0 to event A . Thus, $q_1^1 = Q_1^1(\omega_3) = 1$ and $q_2^1 = Q_2^1(\omega_3) = 1/2$.
- Player 2's announcement does not give new information to Player 1, but Player 1's announcement that the probability of A is 1 informs Player 2 that the true state of the world is ω_3 , because if the true state of the world were ω_1 or ω_2 (the other two possible states of the world according to the current information of Player 2), Player 1 would have announced that he ascribed conditional probability $1/3$ to A . Therefore, $q_1^2 = Q_1^2(\omega_3) = 1$ and $q_2^2 = Q_2^2(\omega_3) = 1$. This is now common knowledge among the players.